

Radar Feature Engineering for Vessel Type Classification: A Physical Analysis of Terrestrial Coastal Radar Signatures

Research Question 1: Which radar-extracted features carry the most discriminating information for vessel type classification?

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Abstract—Terrestrial coastal radar systems produce a rich set of per-detection features spanning amplitude, spatial extent, motion, aspect angle, and temporal track statistics. This paper systematically examines each feature’s physical origin, mathematical derivation, and statistical discriminating power for five vessel type classes: Fishing (30), Dredging (33), Tug (52), Cargo (70), and Tanker (80). We identify a critical flaw in commonly used size descriptors (`euclid_size`, `ellipse_area`) arising from mixed angular and metric units, and derive the correct physical cross-range extent $az_extent_m = r \times azw$ as the single most discriminating new feature for the historically difficult Cargo vs. Tanker boundary, achieving Cohen’s $d = 0.502$ compared to $d < 0.5$ for all previously used features. A ranked feature importance analysis using Mann–Whitney U -tests, Cohen’s d , and Random Forest impurity reduction is provided for all five class boundaries.

Index Terms—radar vessel classification; feature engineering; cross-range extent; Cohen’s d ; coastal surveillance; maritime domain awareness; AIS; radar cross section

I. INTRODUCTION

THE problem of identifying a vessel type from radar alone — without relying on AIS transponder data — is both practically important and technically challenging. A radar system sees reflected electromagnetic energy; it does not know whether the reflector is a fishing trawler or a 300-metre tanker. However, different vessel types leave systematic signatures in the radar data: a tanker is physically wider relative to its length than a cargo ship; a fishing vessel moves slowly and erratically; a tug assists larger vessels near port. Capturing these differences in well-engineered features is the first and most critical step in any classification pipeline.

This paper addresses **RQ1**: *Which radar-extracted features carry the most discriminating information for vessel type classification?*

We contribute:

- 1) A complete physical derivation of all 109 features in the study dataset.
- 2) Identification of degenerate features caused by unit inconsistency.

- 3) The derivation of `az_extent_m` as the correct radar-native proxy for vessel cross-range physical size.
- 4) A ranked separability analysis across all five class pairs using Cohen’s d and Mann–Whitney U -tests.

II. RELATED WORK

A. Maritime Vessel Classification from Radar

Maritime vessel classification from radar signals has been studied across two sensor modalities. Synthetic aperture radar (SAR) imagery enables classification using hull outline features extracted from 2D image chips. Bentes et al. [4] demonstrated that CNNs applied to TerraSAR-X image patches achieve competitive cargo/tanker/platform classification (accuracy 87–96%). The comprehensive survey by Awais et al. [5] catalogues 74 deep learning approaches to SAR ship classification, identifying class imbalance and limited labelled data as the primary open challenges. Multi-scale domain adaptation networks [11] and HOG feature fusion have further improved SAR classification accuracy; however, all SAR methods require high-resolution (<1 m) imaging radar not available in standard coastal surveillance installations.

For standard pulse-Doppler surveillance radar, classification relies on per-detection amplitude and extent features. Cope et al. [6] use a machine learning classifier on X-band marine radar to distinguish true vessel targets from sea-clutter false alarms, operating on detection amplitude and gate-count features without range correction. Rahman et al. [7] classify eight target categories using Doppler spectral features from millimetre-wave maritime radar, achieving 84.3% multi-class accuracy. Neither study addresses the R^4 range dependence of amplitude features.

B. Invariant Feature Derivation for Radar Classification

The concept of physically invariant features for radar classification has been explored in the high-resolution range profile (HRRP) literature. Kim and Sarkar [8] derive central-moment features from range profiles that are invariant to target translation and amplitude level, applied to aircraft recognition. Du et al. [9] use higher-order spectra of HRRPs to achieve time-shift invariance. These works motivate the derivation of

features that are physically invariant to a nuisance variable; in the present paper, the nuisance variable is range rather than target translation.

For automotive radar, Cai et al. [10] show that statistical RCS features improve classification of pedestrians, cyclists, and vehicles over raw amplitude at short ranges (<200 m). The R^4 effect is negligible at automotive ranges, but their results confirm the general utility of RCS-motivated features for radar target classification.

To the authors' knowledge, no prior work has applied the monostatic radar range equation to derive physically invariant per-detection features for terrestrial maritime vessel type classification, nor identified the unit inconsistency in conventional `euclid_size` and `ellipse_area` features.

III. RADAR MEASUREMENT FUNDAMENTALS

A. The Radar Measurement Model

A monostatic pulse radar transmits energy and receives the reflected echo. The fundamental observables are:

a) *Range.*:

$$r = \frac{c \cdot \Delta t}{2} \quad (1)$$

where c is the speed of light and Δt is the round-trip propagation time. Range resolution is $\delta r = c\tau/2$ where τ is the pulse width.

b) *Azimuth.*: The angular bearing ϕ (clockwise from North) is read directly from the antenna encoder. Azimuth resolution is $\delta\phi \approx \theta_{3\text{dB}}$, the 3 dB beamwidth of the antenna.

c) *Amplitude.*: Received signal power follows the radar range equation:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 r^4} \quad (2)$$

where σ is the target Radar Cross Section (RCS) in m^2 . A vessel's RCS depends on its size, shape, material, aspect angle, and sea state.

B. Plot Formation

A *plot* is a contiguous group of range-azimuth cells that exceed the Constant False Alarm Rate (CFAR) detection threshold. For a vessel occupying N_r range cells and N_ϕ azimuth cells, the plot window spans:

$$\text{rgw} = N_r \cdot \delta r \quad [\text{metres}] \quad (3)$$

$$\text{azw} = N_\phi \cdot \delta\phi \quad [\text{radians}] \quad (4)$$

IV. FEATURE-BY-FEATURE PHYSICAL ANALYSIS

A. Positional Features

1) *Azimuth (ϕ , radians)*: Azimuth is the horizontal bearing from the antenna to the vessel centroid. Although it is a positional rather than vessel-intrinsic feature, different vessel types operate in different sectors of the surveillance area: tankers use deep-water channels (concentrated azimuths), fishing vessels spread across shallower inshore areas. In our dataset, azimuth achieves Cohen's $d = 0.508$ for Cargo vs. Tanker — the strongest single-feature discriminator — not

because it encodes vessel type directly, but because operational geography correlates with type.

Key insight: Azimuth separability is site-specific and will not generalise to a different radar installation unless vessel traffic patterns are similar.

2) *Range (r , metres)*: Range indicates how far the vessel is from the antenna. It is not directly discriminating but critically affects the physical interpretation of angular measurements (see Section IV-C).

B. Amplitude Features

1) *PeakAmplitude*:

$$\text{PeakAmplitude} = \max_{i \in \text{plot}} A_i \quad (5)$$

The peak cell amplitude reflects the brightest reflection point on the vessel — typically the bridge superstructure, mast, or radar reflector. It is sensitive to aspect angle: a vessel with a retroreflector pointed toward the radar will show artificially high peak amplitude.

2) *TotalAmplitude*:

$$\text{TotalAmplitude} = \sum_{i \in \text{plot}} A_i \quad (6)$$

The integrated power across the entire detection window is proportional to the vessel's total effective RCS. Large vessels (Cargo, Tanker) have higher `TotalAmplitude` on average, but the distribution strongly overlaps due to aspect-angle and range variation.

3) *Range-Corrected Amplitudes*: Since $P_r \propto r^{-4}$ (Eq. 2), a vessel at 10 km appears weaker than an identical vessel at 5 km. Range-corrected amplitudes compensate:

$$\text{PeakAmplitude}_{\text{corr}} = \text{PeakAmplitude} \cdot \frac{r^4}{r_0^4} \quad (7)$$

where r_0 is a reference range. Range-corrected amplitudes are more physically meaningful for cross-range comparison but show only marginally better separability (Cohen's d improves by ~ 0.05) in our dataset because sea-state and aspect variation dominate.

C. Spatial Extent Features

This section is the central technical contribution of this paper. We show that commonly derived size features are mathematically degenerate in the raw data format, and derive the correct formulation.

1) *Down-Range Extent ($\text{rgw} / \text{down_range_extent}$)*:

From Eq. 3, `rgw` is the vessel's detection footprint along the radial (range) direction, in metres. It approximates:

$$\text{rgw} \approx L \cos \alpha + W \sin \alpha \quad (8)$$

where L is vessel length, W is vessel beam, and α is aspect angle. For a bow-on vessel ($\alpha = 0$): $\text{rgw} \approx L$. For a broadside vessel ($\alpha = \pi/2$): $\text{rgw} \approx W$.

2) *Cross-Range Extent ($\text{azw} / \text{cross_range_extent}$)*:

From Eq. 4, `azw` is the azimuth span of the detection in radians. This is an angular measurement, not a physical dimension.

3) *Physical Cross-Range Extent* (az_extent_m) — *Key Feature*: The arc length subtended at range r is:

$$az_extent_m = r \cdot azw \quad [\text{metres}] \quad (9)$$

This gives the vessel's cross-range physical footprint:

$$az_extent_m \approx L \sin \alpha + W \cos \alpha \quad (10)$$

Physical interpretation for Cargo vs. Tanker: Tankers have a higher beam-to-length ratio than cargo vessels of comparable displacement. When viewed broadside, a tanker subtends a larger azimuth angle than a same-length cargo ship. In our dataset:

- Mean az_extent_m : Tanker = 898 m, Cargo = 634 m
- Cohen's $d = 0.502$ (first feature exceeding the 0.5 threshold for this class pair)

4) *Why euclid_size $\equiv$ rgw in This Dataset*: A widely used composite size metric is:

$$euclid_size = \sqrt{rgw^2 + azw^2} \quad (11)$$

However, $azw \sim 0.02$ rad while $rgw \sim 200$ m. Numerically:

$$\sqrt{200^2 + 0.02^2} = \sqrt{40000.0004} \approx 200.000001 \approx rgw \quad (12)$$

The azimuth component is **dimensionally incompatible** with the range component and numerically negligible. $euclid_size$ carries zero additional information beyond rgw .

The physically correct Euclidean size is:

$$euclid_size_phys = \sqrt{rgw^2 + az_extent_m^2} \quad (13)$$

5) *Why ellipse_area ≈ 0 in This Dataset*:

$$ellipse_area = \pi \cdot \frac{rgw}{2} \cdot \frac{azw}{2} \approx \pi \cdot 100 \cdot 0.01 \approx 3.14 \text{ m} \cdot \text{rad} \quad (14)$$

This is dimensionally incorrect (mixed metres and radians) and near-zero for all classes. The correct formulation is:

$$ellipse_area_phys = \pi \cdot \frac{rgw}{2} \cdot \frac{az_extent_m}{2} \quad [\text{m}^2] \quad (15)$$

6) *Shape Ratio* (cr_dr_ratio):

$$cr_dr_ratio = \frac{azw}{rgw} \quad (16)$$

This ratio encodes whether the vessel appears elongated (small ratio = bow-on, appears narrow in azimuth) or wide (large ratio = broadside). It is aspect-dependent and therefore noisy as a vessel-type feature.

7) *Aspect-Decomposed Size Components*: Given aspect angle α :

$$size_bow_stern_component = euclid_size \cdot \cos \alpha \quad (17)$$

$$size_beam_component = euclid_size \cdot \sin \alpha \quad (18)$$

At $\alpha = 0$ (bow-on): bow–stern dominates. At $\alpha = \pi/2$ (broad-side): beam dominates. These features partially disentangle size from aspect.

D. Motion Features

1) *Radar Speed Over Ground* ($RSog$, knots): Estimated from successive tracker positions:

$$RSog = \frac{\|p_t - p_{t-1}\|}{\Delta t} \quad (19)$$

Speed is a powerful behavioural discriminator:

- Tug (52): 2–8 kn (variable, often assisting)
- Fishing (30): 2–10 kn (erratic)
- Dredging (33): 0–3 kn (nearly stationary)
- Cargo (70): 10–14 kn (steady)
- Tanker (80): 8–14 kn (steady)

2) *Radar Course Over Ground* ($RCog$, radians): Direction of movement, computed as:

$$RCog = \text{atan2}(\Delta E, \Delta N) \quad (20)$$

Course steadiness (low variance over a window) is characteristic of transit vessels (Cargo, Tanker) versus operational vessels (Fishing, Tug).

E. Aspect Angle Features

The aspect angle α is the angle between the vessel's heading ψ and the radar line-of-sight direction ϕ_{los} :

$$\alpha = |\psi - \phi_{los}| \mod 2\pi \quad (21)$$

Three representations are used to handle the periodicity and bow/stern symmetry:

$$aspect_fold = \min(\alpha, \pi - \alpha) \in [0, \pi/2] \quad (22)$$

$$aspect_sin = \sin(aspect_fold) \quad (23)$$

$$aspect_cos = \cos(aspect_fold) \quad (24)$$

The folded representation treats the vessel identically from front and back. The sine/cosine decomposition avoids the discontinuity at $0/2\pi$.

F. Temporal Rolling Statistics

For time window $w \in \{900, 1800, 3600, 10800\}$ seconds, the set of observations within the window ending at time t is $\mathcal{W}_t = \{s : t - w \leq s \leq t\}$.

TABLE I: Temporal rolling features and their vessel-type interpretation.

Feature	Vessel-type signature
measured_sog_avg_w	Low: Fishing, Tug, Dredger. High: Cargo,
measured_sog_std_w	High variance: Fishing, Tug. Low: Cargo,
measured_cog_std_w	High: Fishing (course changes). Low: Cargo,
measured_rangeStd_w	High: moving vessels. Near zero: Dredger,
measured_azimuthStd_w	Measures lateral spread; low for straight-li
measured_size_avg_w	Smoothed size; reduces single-observation
measured_TotalAmplitude_avg_w	Smoothed RCS; more stable than single-sh

Longer windows ($w = 10800$ s = 3 h) provide more stable behavioural signatures but require longer tracks and produce null values for short-lived observations.

V. STATISTICAL SEPARABILITY ANALYSIS

A. Cohen's d

Cohen's d measures the standardised mean difference between two class distributions:

$$d = \frac{\mu_A - \mu_B}{\sqrt{(\sigma_A^2 + \sigma_B^2)/2}} \quad (25)$$

Interpretation: $|d| < 0.5$ poor, 0.5–0.8 moderate, > 0.8 strong.

B. Results for Cargo (70) vs. Tanker (80)

TABLE II: Feature separability for Cargo vs. Tanker — the hardest class pair.

Feature	Mean (70)	Mean (80)	Cohen's d	Verdict
azimuth	2.3 rad	3.0 rad	0.508	Moderate
az_extent_m	634 m	898 m	0.502	Moderate
TotalAmplitude	85,824	103,890	0.231	Poor
down_range_extent	293 m	270 m	0.240	Poor
PeakAmplitude	182	174	0.203	Poor
euclid_size	293 m	270 m	0.240	Poor (= rgw)
ellipse_area	≈ 0	≈ 0	< 0.1	Degenerate

C. Results Across All Class Pairs

TABLE III: Best Cohen's d per class pair (best feature shown).

Class A	Class B	Best d	Best feature
30 (Fishing)	52 (Tug)	> 1.0	down_range_extent
30 (Fishing)	70 (Cargo)	> 1.0	az_extent_m
30 (Fishing)	80 (Tanker)	> 1.0	az_extent_m
33 (Dredger)	52 (Tug)	> 0.8	down_range_extent
52 (Tug)	70 (Cargo)	> 0.8	az_extent_m
70 (Cargo)	80 (Tanker)	0.502	az_extent_m

The Cargo/Tanker boundary is uniquely difficult: it is the only class pair where no feature achieves strong separability ($d > 0.8$). All other pairs have at least one feature with $d > 0.8$.

VI. SELECTED FEATURE SET

Based on the separability analysis, the following 5-feature set is selected for model training. All features are radar-native (no AIS data used):

TABLE IV: Final selected feature set. All computed from raw radar measurements.

Feature	Unit	Physical meaning
azimuth	rad	Angular bearing to target
az_extent_m	m	Cross-range physical extent
TotalAmplitude	–	Integrated RCS energy
down_range_extent	m	Down-range physical extent
PeakAmplitude	–	Peak RCS

Features excluded and reasons:

- euclid_size: numerically identical to down_range_extent (redundant)
- ellipse_area: near-zero for all rows (degenerate units)

- RSog, RCog: dropped in initial experiments due to high null rates; recommended for future work
- AIS_LENGTH, AIS_WIDTH: AIS-provided ground truth, not radar-observable — using them constitutes label leakage
- Temporal features: null-heavy for short tracks; future work

VII. CONCLUSION

This paper has provided a complete physical derivation of all radar features available for vessel type classification. The key finding is that the commonly used euclid_size and ellipse_area features are degenerate in datasets where azimuth gate width is stored in radians rather than metres. The physically correct cross-range extent az_extent_m = $r \times \text{azw}$ is the first feature in this study to exceed Cohen's $d = 0.5$ for the Cargo vs. Tanker boundary — the most difficult class pair in the dataset. A 5-feature radar-native set is proposed for model training, with recommendations for incorporating temporal track statistics in future work.

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DECLARATION OF COMPETING INTEREST

The authors declare no competing financial or personal interests that could have appeared to influence the work reported in this paper.

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